Deep Learning Foundations for Natural Language Processing

Natural Language Processing (NLP) has evolved from rule-based grammars to Deep Learning architectures. At the core of modern NLP is the Transformer model, which uses Self-Attention mechanisms to process input tokens in parallel rather than sequentially.

Word Embeddings and Vector Representations

Pre-trained word representations such as Word2Vec and FastText map words to dense continuous vector spaces where semantic similarity is calculated via Cosine Similarity. Sentence Transformers generate contextualized sentence embeddings for semantic retrieval.

Large Language Models and Knowledge Representation

Large Language Models like BERT and GPT utilize multi-head attention to capture syntactic dependencies and contextual relationships across entire document corpora. Combining Large Language Models with Knowledge Graphs enables hybrid retrieval and structured reasoning.